WFLOW-99: Best Practice Summary

> **Series:** WFLOW — Workflows and Alert Notifications | **Notebook:** 99 | **Created:** March 2026 | **Last Updated:** 11/11/2026

Overview

This notebook consolidates every actionable best practice from the WFLOW series (notebooks 01-09 and the WFLOW-94 and WFLOW-95 LABs) into definitive tables organized by category. Each best practice includes the exact setting or value to use, a priority rating, and the source notebook.

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1. Workflow Design & Architecture

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Use Workflows for all new automation	Problem-triggered workflows rather than Classic alerting profiles + problem notifications. No end-of-life date is published for profiles, but management-zone scoping does not carry over: rebuild it with affected-entity tags or a custom DQL matcher	Critical	WFLOW-01
2	Use a simple workflow for a single-destination notification	One task, a restricted action set (no Run JavaScript, no Run workflow) and no workflow-hours cost	Recommended	WFLOW-01, WFLOW-03

#	Best Practice	Recommended Setting/Value	Priority	Source
3	Use a standard workflow only when you need one	Several tasks, branches, loops or JavaScript make a standard workflow, billed in workflow hours for as long as it exists, not per run. Delete experiments you no longer need	Recommended	WFLOW-01, WFLOW-03
4	Deploy before you wait for a trigger	Triggers only start live workflows; a draft runs only from the **Run** button. Select **Deploy**	Critical	WFLOW-01, WFLOW-02
5	Run production workflows as a service user	Every task runs with the actor's permissions; set a service user as the actor of production workflows that several people work on	Critical	WFLOW-01, WFLOW-09
6	Keep workflows small and focused	One trigger → one outcome. In community practice, stay under ~20 tasks and split larger flows; no per-workflow task cap is published	Recommended	WFLOW-01
7	Set explicit task timeouts	Default 60 minutes, max 7 days: `timeout: <seconds>` per task. Distinct from the documented 120 s runtime limit per action (DQL query, JavaScript/HTTP call)	Critical	WFLOW-01, WFLOW-08
8	Design within the documented limits	1,000 event-triggered executions per hour per workflow (beyond that, HTTP 429 throttling); 1,000-character trigger filter; 10 MB input and result; 10,000 workflows per environment. Concurrent executions run *within system capacity*, with no fixed cap published	Recommended	WFLOW-01, WFLOW-09
9	Start with simple notifications	Basic notifications first, then layer in complexity	Recommended	WFLOW-09
10	Run independent notification tasks in parallel	Tasks with no predecessor start in parallel when the trigger fires (Slack + Teams + Email)	Recommended	WFLOW-03
11	Chain dependent tasks with `predecessors`	A task with predecessors waits for all of them; give it a state condition on each (`SUCCESS` , `OK` , `ERROR` , `NOK` , `ANY`)	Recommended	WFLOW-04, WFLOW-08

2. Triggers

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Use the Problem trigger for incident alerts	Trigger type *Problem* (`davis-problem` in an exported workflow); its payload is the `dt.davis.problems` record	Critical	WFLOW-02
2	Use the Davis event trigger only for per-alert reactions	Put the threshold in an anomaly detector (ALERT-02); trigger on the problem, and use the Davis event trigger only when you need per-alert granularity	Critical	WFLOW-02
3	Grant the actor read access to what the trigger watches	`storage:events:read` for Problem and Davis event triggers; `storage:bizevents:read` , `storage:security.events:read` or `storage:system:read` for Event triggers on those tables; plus `storage:buckets:read`	Critical	WFLOW-02

#	Best Practice	Recommended Setting/Value	Priority	Source
4	Use the Schedule trigger for reports and health checks	Fixed time, time interval, or a five-field cron expression (six-field Quartz expressions are not supported)	Critical	WFLOW-02
5	Test with a manual run before deploying	**Run** in the editor works on a draft; `POST /platform/automation/v1/workflows/<id>/run` (platform token with `automation:workflows:run`) runs the live workflow	Critical	WFLOW-02
6	Scope Problem triggers with affected-entity tags	*Include entities with all defined tags* (`entityTagsMatch: all`, `entityTags: {env: [prod]}` in an export). The trigger has no management-zone filter	Critical	WFLOW-02
7	Select only the problem categories you need	Availability, Error, Slowdown, Resource, Custom, Monitoring unavailable, Info (there is no Performance category); add the **Severity** filter to drop lower levels (WFLOW-04 §3). Every selection uses part of the 1,000-character filter budget	Recommended	WFLOW-02, WFLOW-04
8	Enable Wait for root cause analysis	Advanced options. Dynatrace marks it *Recommended* to avoid triggering on incomplete problem data	Recommended	WFLOW-02
9	Use Minimum duration to suppress short-lived problems	5, 10, 15, 30, 60, 120, 240, 1440 or 10080 minutes; the replacement for an alerting profile's delay. A problem that closes sooner never starts the workflow	Recommended	WFLOW-02, WFLOW-04
10	Handle all problem lifecycle events	Problem state *active or closed*; branch on `event.status` (`ACTIVE` / `CLOSED`) in a single workflow	Recommended	WFLOW-05
11	Validate the filter with Query past events	Zero matches across every window in an environment you know is busy means the filter is over-constrained; nothing else reports it	Recommended	WFLOW-02
12	Stagger scheduled workflows	Start minutes such as `:00`, `:03`, `:07` instead of pinning every workflow to `:00`	Recommended	WFLOW-02
13	Always set a time zone on schedule triggers	`timezone: America/New_York`; a cron schedule otherwise takes the browser's locale	Recommended	WFLOW-02
14	Use the Event trigger for business process automation	Event type `bizevents` with a DQL-matcher filter query (`event.type == "com.company.order-completed"`); send business events as flat JSON	Optional	WFLOW-02

3. Connections & Credentials

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Use descriptive connection names	Pattern: `<service>-<environment>-<purpose>` (e.g., `slack-prod-oncall`)	Critical	WFLOW-03
2	Separate connections per environment	Distinct connections for prod vs staging vs dev	Critical	WFLOW-03
3	Apply least-privilege scopes	Slack bot: `chat:write`, `channels:read` and `groups:read` (Dynatrace's minimal manifest); add `chat:write.public` or `channels:join` only to post to channels the bot has not joined	Critical	WFLOW-03, WFLOW-09
4	Allow every destination host	A host pattern under **Settings > General > External requests** for each connector and HTTP target, and the connector permissions under **Workflows > Settings > Authorization settings**	Critical	WFLOW-03
5	Rotate credentials with a dual-credential overlap	Cadence set by your security policy (community baseline: quarterly for Slack bot tokens and ServiceNow OAuth secrets, annually for PagerDuty integration keys). Mint the new credential, create a `-v2` connection, migrate workflows, then revoke the old credential	Recommended	WFLOW-09
6	Prefer OAuth Client Credentials for ServiceNow	The connection accepts only Basic Authentication or OAuth Client Credentials. Use a dedicated integration user with table-level rights, never `admin`	Recommended	WFLOW-05, WFLOW-09
7	Pick the PagerDuty connection by action	An Events connection (routing key) for **Send event**; a REST API key connection for **Create an incident** and the list actions. Separate Events connections per routing target (`pagerduty-prod-critical`, `pagerduty-prod-standard`, `pagerduty-platform`)	Recommended	WFLOW-05
8	Use a bot token for the Slack connection	The Slack connector takes a Slack app bot token and has no webhook option. An incoming webhook is reachable only from an HTTP Request task	Recommended	WFLOW-03

4. Notification Channels

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Multi-channel for Critical production problems	`event.severity` 1: PagerDuty + Slack #alerts-urgent + Email simultaneously	Critical	WFLOW-03, WFLOW-04
2	Slack + email for Major production problems	`event.severity` 2: Slack #alerts-production + Email	Recommended	WFLOW-04
3	Slack only for Minor problems	`event.severity` 3, or unset: Slack #alerts-production; staging gets a daily digest. Problems never carry 4 or 5	Recommended	WFLOW-04
4	Use a Teams Workflows (Power Automate) webhook	Office 365 connectors stopped working in May 2026. The Teams connection takes the URL from *Send webhook alerts to a channel*; allow its host (e.g., `*.api.powerplatform.com`) under External requests	Critical	WFLOW-03

#	Best Practice	Recommended Setting/Value	Priority	Source
5	Use the built-in Send email action for simple notifications	Sends from `no-reply@apps.dynatrace.com`, up to 10 addresses per To/Cc/Bcc field, markdown rather than HTML, and needs `email:emails:send`. It has no SMTP option: for a corporate sender or high volume, call the provider's API (SendGrid, SES, a relay) from an HTTP Request task	Optional	WFLOW-03
6	Include the problem category in the email subject	`[{{ event()['event.category'] }}] {{ event()['event.name'] }}`	Recommended	WFLOW-03

5. Message Formatting & Templates

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Structure messages: Severity + Title + Key metrics + Link	Line 1: severity indicator + title; Line 2: impact; Line 3: affected entities; Line 4: action link	Critical	WFLOW-06
2	Use severity-coded visual indicators	Slack, keyed on `event.severity`: `:red_circle:` 1 Critical, `:large_orange_circle:` 2 Major, `:large_yellow_circle:` 3 Minor, `:large_blue_circle:` 4 Warning (Davis events only; problems carry 1–3)	Critical	WFLOW-06
3	Use Slack Block Kit for rich messages	Block Kit JSON in `message` with `messageFormat: slack_format`; the action has no separate `blocks` input. A double quote in a resolved value breaks the JSON, so send a test message	Recommended	WFLOW-03, WFLOW-06
4	Use Teams Adaptive Cards v1.4	Card JSON (`type: AdaptiveCard`, `version: "1.4"`, `FactSet`, `TextBlock`, `Action.OpenUrl`) in `message` with the default `msteams_format`, or a `selectTemplate` card. Use Dynatrace expressions; Adaptive Card templating is not supported	Recommended	WFLOW-03, WFLOW-06
5	Always include a link to the Dynatrace problem	`<{{ problem_link() }}|View in Dynatrace>` (Slack) or `Action.OpenUrl` (Teams). `problem_link()` evaluates only under a Problem trigger	Critical	WFLOW-03
6	Use dictionary mapping for severity-to-emoji	`{{ {1: ":red_circle:", ...}.get(event().get("event.severity") | int(5), ":white_circle:") }}`	Recommended	WFLOW-06
7	Limit affected entities shown to 3-5	Read Smartscape names first and fall back to the deprecated classic IDs, which are better populated today: `(event().get("smartscape.affected_entities") or []) | map(attribute="name") | join(", ") or event()["affected_entity_ids"] | join(", ")`. Slice to `[:3]` with an overflow indicator	Recommended	WFLOW-05, WFLOW-06

#	Best Practice	Recommended Setting/Value	Priority	Source
8	Read optional fields with `.get()`	`(event().get("root_cause.smartscape_entity") or {}).get("name", "not yet determined")`. Most problems name no root cause (79% had neither `root_cause.smartscape_entity` nor the deprecated `root_cause_entity_id` over 7 days on a validation tenant, 10/06/2026), and a bracket read of a missing field fails the task with *Undefined variables*	Critical	WFLOW-02, WFLOW-03, WFLOW-07
9	Truncate long strings	`{{ event()["event.name"] | truncate(50) }}`	Recommended	WFLOW-06
10	Enrich notifications with DQL data	A JavaScript task before the notification task runs `queryExecutionClient.queryExecute()` and polls with `queryPoll()` until the query finishes; skip it when the problem names no root cause	Optional	WFLOW-06, WFLOW-08
11	Test templates before production	An On-Demand workflow with a JavaScript mock event using the `dt.davis.problems` field names. `problem_link()` renders only under a Problem trigger, so confirm the link on a real problem. A template that fails to evaluate shows up as `Error evaluating` on the `WORKFLOW_EXECUTION` record	Critical	WFLOW-06
12	Preview Slack templates in Block Kit Builder	https://app.slack.com/block-kit-builder before deploying	Recommended	WFLOW-06
13	Create a resolved-problem template	`:white_check_mark: Problem Resolved` with the duration from `resolved_problem_duration` and the resolution time from `event.end`	Recommended	WFLOW-06

6. Routing & Escalation

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Route by severity	`event.severity` 1 Critical: PagerDuty + Slack + Email; 2 Major: Slack + Email; 3 Minor, or unset: Slack. Problems never carry 4 (Warning) or 5 (Informational). Check which sources set severity: SaaS 1.348 stops defaulting it to 3 (rollout start 10/12/2026, staged tenant rollout)	Critical	WFLOW-04
2	Route by team ownership tags	Preferred: one workflow per team, whose Problem trigger filters on the built-in ownership tag (`owner` / `dt.owner`) with the *Affected entities* tag filter, so the match is made before any task runs. Fallback in a task condition: problem records carry tags in `entity_tags` (an array of `key:value` strings, deprecated), so `"owner:checkout" in (event().get("entity_tags") or [])` is an exact element test. A `tags` field read with `.get("tags", "")` no longer appears on problem records and always evaluates false	Critical	WFLOW-04
3	Do **not** route by management zone	`"Production" in event()["management_zones"]` is legacy, and the field is not present on problem records (0 of 15,878 over 30 days on a live tenant, 10/05/2026, with management zones still in use), so the condition has nothing to match. Route on ownership tags or Smartscape ownership instead	Critical	WFLOW-04

#	Best Practice	Recommended Setting/Value	Priority	Source
4	Implement time-based routing	Business hours: Slack channel; off-hours Critical: PagerDuty; off-hours non-critical: queue for morning. Pass the team's time zone, `now("America/New_York")`; `now()` is UTC otherwise. Business calendars (`calendars()`) cover holidays	Critical	WFLOW-04
5	Escalate problems that are still active	Task option **Wait before** (`waitBefore: 900`), then re-query `dt.davis.problems` and escalate to PagerDuty if `event.status` is still `ACTIVE`; or a second workflow with Minimum duration 15. Neither sees acknowledgment: leave acknowledgment-aware escalation to the paging tool	Recommended	WFLOW-04
6	Multi-tier escalation	0 min: Slack; 15 min: Email team lead; 30 min: PagerDuty on-call; 60 min: PagerDuty manager	Recommended	WFLOW-04
7	Combine checks in one custom condition	Each task has one `conditions` block: `states` of its predecessors, one `custom` Jinja expression with `and` / `or` / `not` inside it, and `else: SKIP` on branches that may not run (the default is stop). E.g. `custom: '{{ (event().get("event.severity") | int(5)) <= 1 and "env:prod" in (event().get("entity_tags") or []) }}'`	Recommended	WFLOW-04
8	Use JavaScript for dynamic channel selection	Read the `owner:` tag from `entity_tags` (deprecated; may arrive as an array or a JSON string) to build `#${team}-alerts`, or use the Ownership app's **Get owners** action for each team's Slack channels	Optional	WFLOW-04, WFLOW-08
9	Separate environment routing	Production: immediate notification; Staging: Slack only; Dev: daily digest	Recommended	WFLOW-04

7. Incident Management Integration

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Deduplicate PagerDuty incidents	`dedupKey: "dynatrace-{{ event()['display_id'] }}"` on every Send event for the problem, through the same Events connection	Critical	WFLOW-05
2	Deduplicate ServiceNow incidents	Set `correlationId: "DT-{{ event()['display_id'] }}"` and run **Search incidents** on it before **Create Incident**, creating only when nothing is found. Create Incident is a plain POST and does not deduplicate on the correlation ID	Critical	WFLOW-05
3	Map Dynatrace severity to PagerDuty severity	`event.severity` 1: `critical`; 2: `error`; 3: `warning`; 4: `info`	Critical	WFLOW-05
4	Map Dynatrace severity to ServiceNow impact/urgency	`event.severity` 1–2: impact=1, urgency=1; 3: impact=2, urgency=2; 4: impact=3, urgency=3; 5: usually no incident	Critical	WFLOW-05
5	Auto-resolve incidents when the problem closes	Problem state *active or closed*. On `event.status == "CLOSED"`: Send event with `eventAction: resolve` and the same `dedupKey` (PagerDuty), or **Resolve incident** with the incident number from Search incidents, resolution notes and a resolution code from your instance's list (ServiceNow)	Critical	WFLOW-05

#	Best Practice	Recommended Setting/Value	Priority	Source
6	Include Dynatrace problem URL in incident details	`{{ problem_link() }}` in `customDetails` (PagerDuty) or the `description` body (ServiceNow)	Critical	WFLOW-05
7	Write the incident number back onto the problem	Read `result("create_snow_incident").number` downstream and post it with `problemsClient.createComment()` on `event()["event.id"]`. Comment and Resolve take the incident number, not the `sys_id`. Deprecated in Dynatrace API 1.349 (staged rollout from 10/27/2026): *"POST /problems/{problemId}/comments Deprecated"*, no successor named	Recommended	WFLOW-05
8	Use PagerDuty Events API v2	Integration type: Events API v2 (not v1)	Critical	WFLOW-05
9	Fill ServiceNow's required fields	Category, Subcategory, Impact, Urgency and Assignment Group are required on Create Incident. Pick the assignment group in the editor (it is stored as the group's sys_id) and set Caller	Critical	WFLOW-05
10	Implement bi-directional sync	A scheduled workflow polls ServiceNow for open DT-correlated incidents, with a Credential Vault credential, and writes their status back as problem comments; or the ServiceNow-side Dynatrace app / ITOM (ALERT-04)	Optional	WFLOW-05
11	Harden the integration	**Retry on error** for transient failures (fixed count and delay, no backoff, retries every failure); route the task's error state to a dead-letter task; monitor connector `ACTION_EXECUTION` errors	Recommended	WFLOW-05

8. Auto-Remediation

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Rate-limit remediation attempts	Max 3 remediation runs per hour, counted with `countDistinctExact(dt.automation_engine.workflow_execution.id)` excluding the current run. The check task throws at the limit, so a remediation task gated on its `SUCCESS` does not run. The count spans all entities; a per-entity limit needs the entity recorded, for example in a business event	Critical	WFLOW-07
2	Enforce time-window guardrails	A `custom` condition on the remediation task, weekdays 06:00–22:00 in a named time zone (`now("Europe/Vienna")`), with `else: SKIP`	Critical	WFLOW-07
3	Implement cooldown periods	30 minutes minimum between remediation actions on the same entity	Critical	WFLOW-07
4	Capture pre-remediation state for rollback	Record current state before executing any change	Critical	WFLOW-07

#	Best Practice	Recommended Setting/Value	Priority	Source
5	Gate every action on a root-cause entity	Read `root_cause.smartscape_entity` first, fall back to the deprecated `root_cause_entity_id`, and end with a "skipped" result when neither is present. Most problems name none (79% over 7 days on a validation tenant)	Critical	WFLOW-07, WFLOW-08
6	Require human approval for production remediation	Slack **Request approval** action (copy its ID from the action picker) with a 30-minute task timeout; test what a Decline and a timeout leave before gating on it. Auto-approve only when a primary tag positively marks the problem non-production, so a missing tag fails closed to approval	Critical	WFLOW-03, WFLOW-07
7	Auto-remediate only well-defined scenarios	Pod crash loops: yes (restart the workload); Disk space 90%: yes (cleanup); Database deadlocks: no (investigate); Security incidents: no (human judgment)	Critical	WFLOW-07
8	Remediate through connectors	Kubernetes Connector **Rollout restart resource** on the one workload the problem names (it has no rollback action; leave rollback to your CD tooling), and AWS Connector **Reboot instances** after resolving the instance ID from Smartscape. A Run JavaScript task cannot import npm client libraries by name	Recommended	WFLOW-07
9	Follow remediation maturity model	Level 0: manual; Level 1: notification + runbook link; Level 2: semi-automated (approval); Level 3: fully automated with guardrails; Level 4: self-healing with learning	Recommended	WFLOW-07
10	Map problem types to runbooks	JavaScript `runbookMap` object mapping problem title patterns to runbook IDs	Recommended	WFLOW-07
11	Include runbook links in notifications	Jinja conditional: `{% if "CPU" in event()["event.name"] %}` links to CPU runbook	Recommended	WFLOW-07
12	Validate after remediation	Query metrics/logs post-action to confirm resolution	Recommended	WFLOW-07
13	Start non-prod, promote to prod	Test all remediation workflows in staging before enabling in production	Critical	WFLOW-07

9. JavaScript & HTTP Actions

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Always use `export default async function`	The runtime injects only `executionId` and `actionExecutionId`; read the event with `execution()` and predecessor results with `await result()` from `@dynatrace-sdk/automation-utils`	Critical	WFLOW-08

#	Best Practice	Recommended Setting/Value	Priority	Source
2	Handle errors deliberately	`try` / `catch` for recoverable errors, returning a degraded result. Throw when the failure is terminal, so the task ends in `ERROR` and an on-failure task can route around it; a guardrail that returns `{proceed: false}` leaves its task in `SUCCESS`	Critical	WFLOW-07, WFLOW-08
3	Retry only what can succeed	Max 3 attempts, delay = `1000 * attempt` ms, on network errors and 5xx only (a 4xx is thrown at once). Retry a `POST` only if the target treats a repeat as a no-op	Recommended	WFLOW-08
4	Set request timeouts on external HTTP	`AbortController` with a 10-second timeout on outbound `fetch()` calls. Distinct from the task `timeout` (whole-task budget) and the 120 s runtime budget (per action). See WFLOW-08 §9–§10	Critical	WFLOW-08
5	Poll DQL queries to completion	`queryExecute()` returns the result only if the query finishes within `requestTimeoutMilliseconds`; otherwise it returns a `requestToken`. Poll with `queryPoll()` until the state is final, and fail on anything but `SUCCEEDED` — never read `result.result.records` unchecked. The 120 s runtime budget still applies. See WFLOW-08 §2 and §10	Critical	WFLOW-08
6	Limit DQL query scope	`from: now() - 1h`, select only needed `fields`, `limit 100`	Critical	WFLOW-08
7	Use `Promise.all()` for parallel entity lookups	Execute all independent API calls concurrently	Recommended	WFLOW-08
8	Always use HTTPS for external calls	Never use `http://` in HTTP request URLs	Critical	WFLOW-09
9	Allow-list every outbound host	Add each host a Run JavaScript or HTTP Request task calls under **Settings > General > External requests**; a call to an unlisted host fails with `Blocked request to '…' (host not in allowlist)`. Hosts on a private network, and targets that admit only allow-listed source IPs, are reached through EdgeConnect (WFLOW-94)	Critical	WFLOW-08, WFLOW-94
10	Read secrets from the Credential Vault	`credentialVaultClient.getCredentialsDetails({ id })` in JavaScript (actor has `credential-vault:entries:read`, credential scoped to AppEngine, *Allow access without app context* on, actor granted access); the Authentication field in HTTP tasks. Never hardcode, log or return a secret	Critical	WFLOW-08, WFLOW-09
11	Give each step its own failure task	A task waits for all of its predecessors, so one failure task chained after several steps never fires. Give each step a failure task with a single predecessor set to `NOK` and `else: SKIP`	Recommended	WFLOW-08
12	Add problem comments after remediation	`problemsClient.createComment()` with the problem ID `event.id`, not `display_id`. Deprecated in Dynatrace API 1.349 (staged rollout from 10/27/2026): *"POST /problems/{problemId}/comments Deprecated"*, no successor named	Recommended	WFLOW-08
13	Validate secret existence before use	`if (!apiToken) throw new Error('Missing required secret: ...')`	Recommended	WFLOW-09

#	Best Practice	Recommended Setting/Value	Priority	Source
14	Grant the workflow the scopes its tasks use	**Workflows > Settings > Authorization settings** (for example `environment-api:credentials:read`, `environment-api:problems:write`, the `storage:*:read` of each queried table); a missing scope fails the task with a 403	Recommended	WFLOW-07, WFLOW-08

10. Security

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Never hardcode secrets in workflows	Connections for connector actions; Credential Vault for HTTP (Authentication field) and JavaScript (`credentialVaultClient`)	Critical	WFLOW-09
2	Sanitize all dynamic inputs	Remove `"'\\` characters, limit string length to 200 chars before using in queries	Critical	WFLOW-09
3	Never log secrets	Do not `console.log()` a credential or return it in task output — results are visible to anyone who can read the workflow	Critical	WFLOW-09
4	Name credentials consistently	Pattern: `<service>-<env>-<purpose>` for Credential Vault entries and connections	Recommended	WFLOW-09
5	Rotate secrets on your policy's cadence	Dual-credential overlap: mint the new credential, create a `-v2` connection, migrate workflows one at a time, wait a full execution cycle, then revoke the old one at its source. On suspected exposure, revoke first	Recommended	WFLOW-09
6	Fail secure on errors	Default to safe state (no action) when errors occur, never expose error details externally	Critical	WFLOW-09

11. Access Control

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Grant minimum permissions per role	Viewers: `automation:workflows:read`; builders: `read` + `write` + `run`; workflow admins add `automation:workflows:admin` (Workflow admin mode). Using Workflows needs `app-engine:apps:run`, and writing and running them `app-engine:functions:run`. Which workflows a user sees follows ownership and visibility, not a policy condition	Critical	WFLOW-01, WFLOW-09
2	Limit who can create billed workflows	`ALLOW automation:workflows:write WHERE automation:workflow-type = "SIMPLE";` for app teams. `automation:workflow-type` is the only condition on this permission; IAM has no workflow-owner condition	Recommended	WFLOW-09
3	Separate connection write access	Connections are settings objects: grant `settings:objects:write` on the connector's schema (e.g., `WHERE settings:schemaId = "app:dynatrace.pagerduty:connection"`) only to workflow admins and SRE. There is no `automation:connections:*` permission	Critical	WFLOW-09
4	Record ownership in the workflow's own fields	`owner` with `ownerType: GROUP` for team ownership, `isPrivate`, and a service-user `actor`; team and classification in the `description` or the naming convention. There is no free-form `metadata` ownership block	Recommended	WFLOW-09

#	Best Practice	Recommended Setting/Value	Priority	Source
5	Control who can assign service-user actors	`ALLOW iam:service-users:use WHERE iam:service-user-email IN ("<SERVICE_USER_EMAIL>");` . A person who edits a workflow becomes its actor unless the actor is a service user or the edit is made in Workflow admin mode	Critical	WFLOW-09

12. Observability & Monitoring

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Alert on workflow success rate	Threshold: < 95% success rate over 7 days	Critical	WFLOW-09
2	Alert on average execution duration	Threshold: > 5 minutes average	Recommended	WFLOW-09
3	Alert on failure spike	Threshold: > 5 failures per hour	Critical	WFLOW-09
4	Build a workflow health dashboard	Queries on `dt.system.events` (`event.kind == "WORKFLOW_EVENT"`): overall health, per-workflow success rates, execution trend, recent failures, task-level performance. Count final records only (`dt.automation_engine.state.is_final == true`)	Critical	WFLOW-09
5	Create a workflow-monitoring workflow	Scheduled; query for workflows with 3+ failures in the last hour; alert to `#workflow-alerts` . Its actor needs `storage:system:read` and `storage:buckets:read`	Recommended	WFLOW-09
6	Monitor notification task success rates	Query final `ACTION_EXECUTION` records in `dt.system.events` by `dt.automation_engine.action.app` (`dynatrace.email` , `dynatrace.slack` , …)	Recommended	WFLOW-03
7	Track remediation success rates	Query `WORKFLOW_EXECUTION` records filtered on `dt.automation_engine.workflow.title` containing `remediation` (`caseSensitive: false`), or on the remediation workflows' IDs	Recommended	WFLOW-07
8	Monitor skipped tasks	Query `TASK_EXECUTION` records with `dt.automation_engine.state` `SKIPPED` / `DISCARDED` to validate routing conditions are working as designed	Optional	WFLOW-04
9	Exclude tasks that never ran from duration statistics	`filter in(dt.automation_engine.state, {"SUCCESS", "ERROR"})` ; `DISCARDED` and `SKIPPED` tasks are final but always carry duration 0	Recommended	WFLOW-09
10	Watch for throttling and blocked hosts	The Workflows overview's throttled filter (1,000 event-triggered executions per hour per workflow); `Blocked request … (host not in allowlist)` in `dt.automation_engine.state_info` means a missing External requests entry	Recommended	WFLOW-09

13. Operations & Change Management

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Name workflows consistently	Pattern: `<team>-<function>-<environment>` (e.g., `sre-problem-notifications-prod`)	Critical	WFLOW-09
2	Export workflows for version control	`GET /platform/automation/v1/workflows/<id>/export` with a platform token (`automation:workflows:read`, sent as `Bearer`), saved to Git	Recommended	WFLOW-09
3	Test changes on cloned workflow first	Clone, modify, test with On-Demand trigger, review results, promote	Critical	WFLOW-09
4	Rollback procedure	Disable the failing workflow, restore the previous version from its history (`POST /platform/automation/v1/workflows/<id>/history/<version>/restore`, which deploys it), enable it, verify. Import your own export only if the history no longer has the version	Critical	WFLOW-09
5	Daily: check execution dashboard	Review success rates and investigate failures	Recommended	WFLOW-09
6	Weekly: verify external connections	Test all Slack, Teams, PagerDuty, ServiceNow connections	Recommended	WFLOW-09
7	Rotate secrets per your policy	Dual-credential overlap, then revoke the old credential (§10 row 5)	Recommended	WFLOW-09
8	Query execution history with DQL	`fetch dt.system.events, from:-24h | filter event.kind == "WORKFLOW_EVENT" and event.type == "WORKFLOW_EXECUTION" | filter dt.automation_engine.state.is_final == true` — without `is_final`, a run is counted once as `RUNNING` and again in its final state	Recommended	WFLOW-01

Summary

This notebook contains **128 best practices** across 13 categories extracted from the complete WFLOW series.

Priority distribution:

Priority	Count	Meaning
Critical	62	Must implement for production readiness
Recommended	60	Strongly advised for operational maturity
Optional	6	Beneficial for advanced use cases

Implementation order:

1. Connections, External requests host patterns and a service-user actor (WFLOW-01, WFLOW-03, WFLOW-09)
2. A simple problem-notification workflow, then severity and ownership routing (WFLOW-01, WFLOW-03, WFLOW-04)
3. Incident management integration with search-before-create deduplication (WFLOW-05)
4. Custom templates with severity indicators and links (WFLOW-06)
5. Workflow health monitoring dashboard (WFLOW-09)
6. Auto-remediation with guardrails (WFLOW-07)

References

- Workflows umbrella (DT docs)
- Workflow triggers (DT docs)
- Workflow actions (DT docs)
- Notification actions umbrella (DT docs)
- HTTP request action (DT docs)
- Run JavaScript action (DT docs)
- Workflow reference / Jinja expressions (DT docs)
- Davis Problems app (DT docs)
- Alerting and notifications umbrella (DT docs)
- Upgrade guide — Alert notification (DT docs)
- Event triggers for workflows (DT docs)
- Build workflows (DT docs)
- Manage workflow permissions (DT docs)
- Automation Workflow consumption (DT docs)
- IAM policy statements (DT docs)
- Set up Slack Connector (DT docs)
- Email (DT docs)
- ServiceNow Connector (DT docs)
- EdgeConnect (DT docs)
- Dynatrace Developer Portal (Dynatrace)

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